



If hardware is the heart of a computer, then the operating system (OS) is its soul. It's the OS that gives life to a computer and makes it perform various tasks. That's why it's important to choose the right OS for a particular computer system. Fuchsia is an open source compatibility based operating system currently being developed by Google, which addresses the challenges the Android OS faces.

Google Fuchsia: Improving on the Android OS

The design and development of an OS plays an important role in the performance of computer hardware, and a perfectly designed OS can make the hardware run at its full potential. Basically, there are seven types of OSs – single- and multi-tasking, single- and multi-user, distributed, templated, embedded, real-time, and library. Among these, the single- and multi-tasking OSs are more widely used in many computer systems. The multi-tasking OS, also known as the General Purpose OS (GPOS), is used because it can do multiple tasks at the same

time. GPOSs in use today include Mac, Windows, Linux, Android and Raspberry Pi. They work on a method called time-slicing, whereby the OS slices the CPU time into various time slots. Each time slot is dedicated to a task, which is scheduled accordingly. This process is also known as pre-emptive multi-tasking. The time scheduling of the task is done in such a way that the system throughput is maintained. However, GPOSs don't give time assurance for the completion of any task, because tasks are not executed based on their priority, and can cause latency.

In the case of a Real Time OS (RTOS), every task is executed according to its priority and can be completed within the expected time. That's why RTOSs don't cause any latency like the GPOSs, and are more responsive to the I/O data. As most smartphones use Android and iOS, the user experiences some kind of latency at some point of time because both the OSs use pre-emptive multi-tasking and are not RTOSs. Moreover, Google has developed the Android OS from the kernel of the Linux OS, which was not designed for smartphone hardware. To compensate for this incompatibility,

